Fundamentals of Cybersecurity

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Network

Wireless Network Security

Wireless communication

- Wireless networks allow devices to connect and communicate without the need for physical cables.
- They use electromagnetic waves, such as radio frequencies, to transmit data over the air.

  - The signals are **available to anyone** with an effective antenna within range.



Wireless communication

- Wireless networks are inherently more vulnerable than wired ones because data is transmitted over the air.
- Here are some risks:
  - Eavesdropping: Attackers can intercept unencrypted data.
  - Unauthorized Access: Without strong security protocols, strangers can connect to your network.
  - Man-in-the-Middle Attacks: Hackers can intercept and manipulate the communication between devices. (*How?*)

The 802.11 protocol suite

- WiFi: a wireless networking technology that allows devices to connect to the internet and communicate wirelessly.
- Key Components
  - **Wireless Access Point** (WAP): Acts as a hub for wireless communication.
  - **Client Devices**: Smartphones, laptops, IoT devices, etc.
  - **Router/Modem**: Connects the local network to the internet.
- Frequency Bands: 2.4 GHz, and 5 GHz
- Data is transmitted over specific channels within the frequency band.
  (CSMA/CD-based time sharing on the same channel)

WiFi frames

- Each WiFi data unit is called a frame.
- Each frame contains three fields: MAC header, payload, and FCS.
  - Some MAC header fields:
    - Frame type: control, management, or data
    - ToDS, FromDS: direction of this frame: to or from the access point (called the distribution system)
    - MAC addresses
    - Fragmentation and order control bits
  - Payload: the actual data being transmitted (0–2304 bytes)
  - FCS (frame check sequence): an integrity check code (CRC)

WiFi frames

- WiFi frames are categorized into three types:
  - Data Frames: Carry the actual user data.
  - Control Frames: Facilitate data transmission (e.g., RTS/CTS, ACK frames).
  - Management Frames: Establish and maintain connections (e.g., beacon frames, probe requests, authentication).

Management frames

The most significant management frame types:

- Beacon: Each access point periodically sends a beacon frame to announce its presence. Any NICs that are within range receive this beacon.
  - It includes SSID, transmission rate, supported security protocols
- Authentication: A NIC initiates a request to interact with an access point by sending its identity in an authentication frame.
  - The access point may either accepts or rejects the request.
- Association: Following authentication, a NIC requests an access point to establish a session.
  - The NIC and access point exchange information about their capabilities and agree on parameters of their interaction, especially the encryption algorithm.

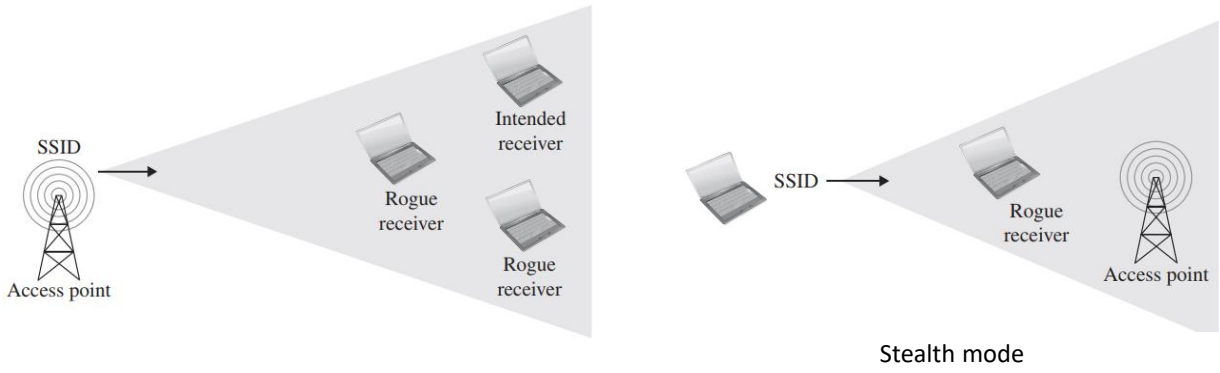
SSID

- SSID (Service Set Identifier): a character string to identify an access point
  - A string of up to 32 characters chosen by the access point's administrator
  - The access point broadcasts in its beacon.
  - SSIDs need to be unique in a given area to distinguish one wireless network from another.

SSID modes

- Normal: an access point continually broadcasts its appeal in its beacon, indicating that it is open for the next step in establishing a connection.
- stealth mode (SSID cloaking): The client must first send a signal seeking an access point with a particular SSID.
  - If you do not know the SSID, you cannot request a connection.
  - However, closed mode leaves the client exposed. In open mode, the client is quiet, monitoring beacons, until it finds one to which it wants to connect. In closed mode, the client effectively becomes a beacon.

WiFi modes



Stealth mode

- From those beacons and messages a rogue host can learn the expected values needed to impersonate an access point to which the client hopes to connect.

MAC-based Authentication

- Access points can manage lists of MAC addresses of devices with which they will accept connections.
  - Thus, authentication (in step 2) could be accomplished by accepting only devices on the positive accept list.
- Vulnerable to MAC spoofing
  - The MAC address of most NICs can be changed via software (OS)
  - One device impersonates another.

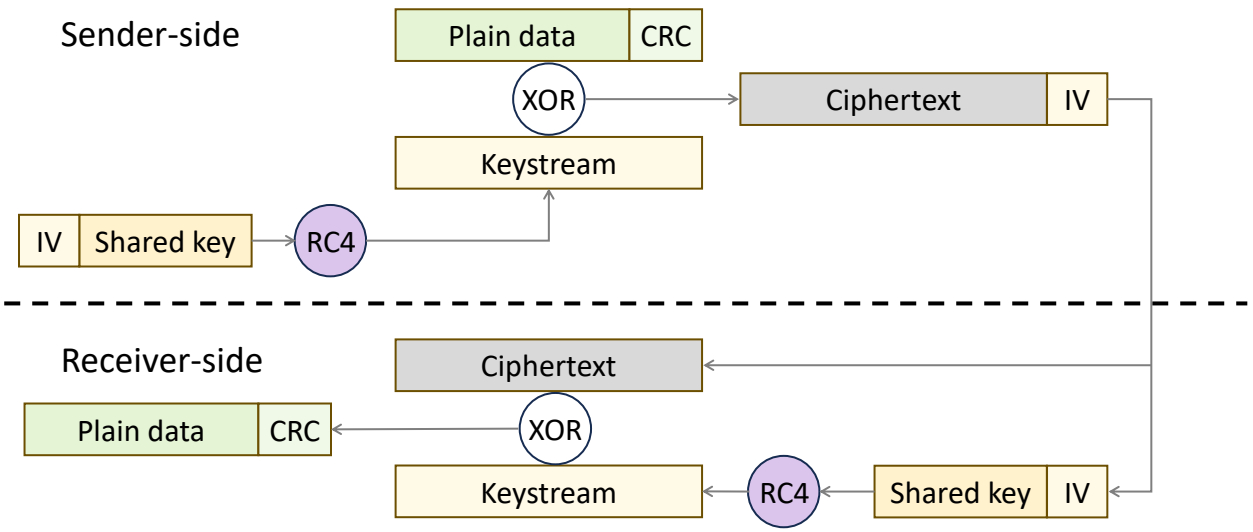
Wireless security protocols

- 1999: WEP (*Wired Equivalent Privacy*) introduced with the original IEEE 802.11 standard.
- 2003: WPA (*Wi-Fi Protected Access*) developed to address WEP's vulnerabilities.
- 2004: WPA2 introduced with stronger encryption standards.
- 2018: WPA3 released with advanced security mechanisms for modern threats.

WEP

- Designed to provide a level of security equivalent to wired networks.
- Uses **RC4 stream cipher** for encryption.
- Requires a shared secret key that is manually configured on both the client and the access point.
- Key lengths are typically 40-bit or 104-bit. The actual key is expanded by appending a 24-bit Initialization Vector (IV).
- The IV is a random sequence of bits appended to the shared key.
- The IV is sent in plaintext along with the encrypted data.
- Authentication: A challenge-response mechanism is used to verify that the client knows the WEP key.

WEP encryption/decryption



Weaknesses of WEP

- The 24-bit IV is too short (about 16M different IV possibilities).
  - With a large number of packets, IVs are reused, leading to IV collisions.
  - Attackers can analyze packets with the same IV to extract information about the shared key
- Lack of dynamic key management and distribution.
  - All devices on the network use the same shared (static) key.
  - If the key is compromised, the entire network is exposed.
- Weak authentication
- The RC4 cipher is vulnerable to certain attacks.
- WEP does not protect against replay attacks.

❖ WEP was officially deprecated by the IEEE in 2004 due to its severe security flaws.

WPA

- WPA has a key change approach, called **Temporal Key Integrity Program** (**TKIP**), by which the encryption key is changed automatically on each packet.
- Each packet includes a sequence number. The access point and client track the sequence number to ensure packets are received in the correct order and prevent replay attacks.
- In WPA, authentication can be done by password, token, or certificate.
- Keys are generated dynamically for each session, ensuring that no two clients share the same encryption key.
- Cryptographic-based integrity check (MIC algorithm)
- WPA2 adds AES as a possible encryption algorithm (RC4 is also still supported to maintain compatibility).